5 A 100 kg crate is pulled from rest across a floor with a constant force of 320 N. For the first 20.0 m, the floor is frictionless and for the next 10.0 m, a constant frictional force of 30.0 N acts on the crate. What is the final speed of the crate?

A 8.00 m s^{-1} **B** 8.37 m s^{-1}

C 13.6 m s^{-1} **D** 13.9 m s^{-1}